

MBAi 448 | Winter 2026

Generative AI

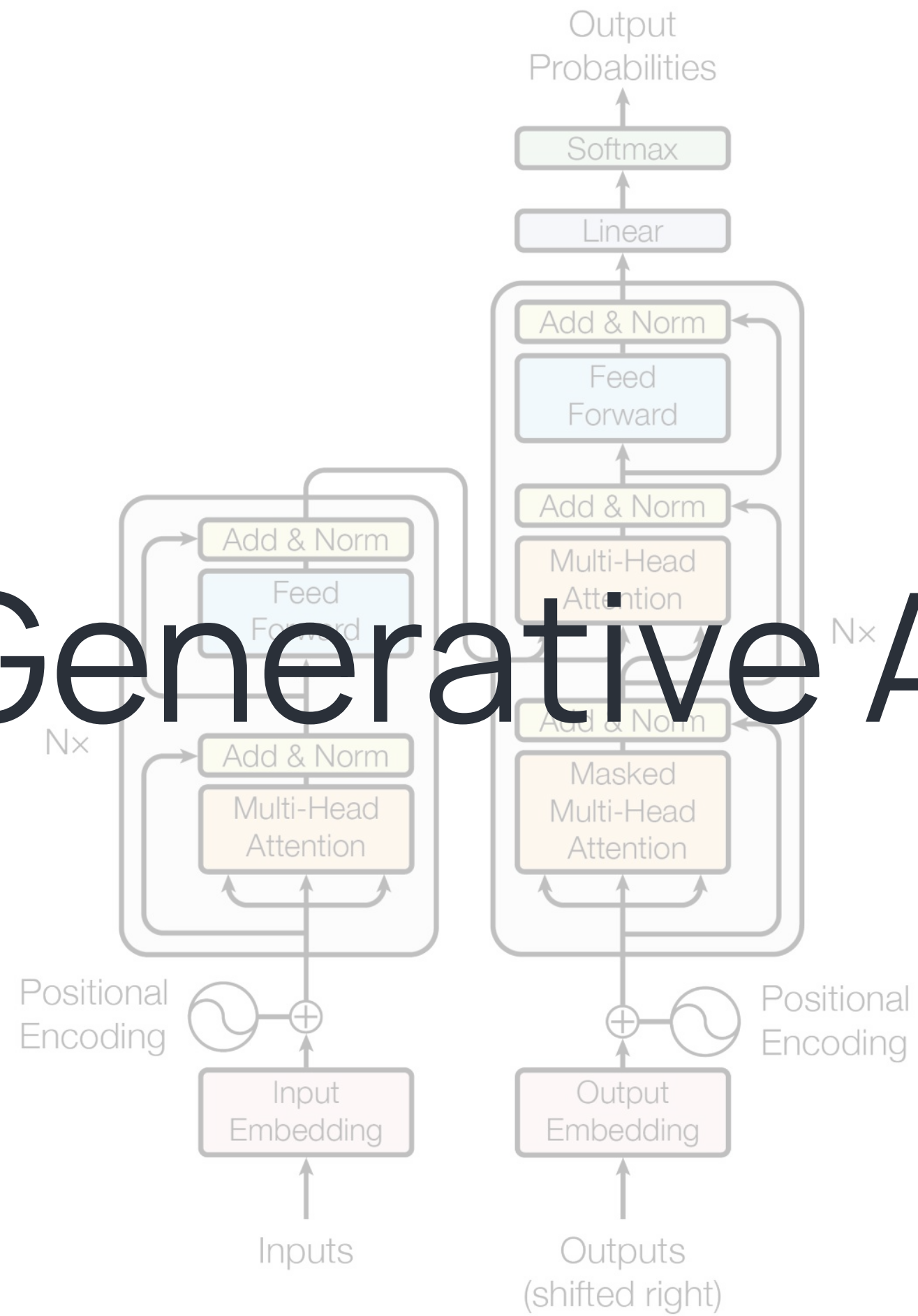
Attendance

Enter the “Magic Word” in Canvas

Do Not Share with Absent Students



Generative AI



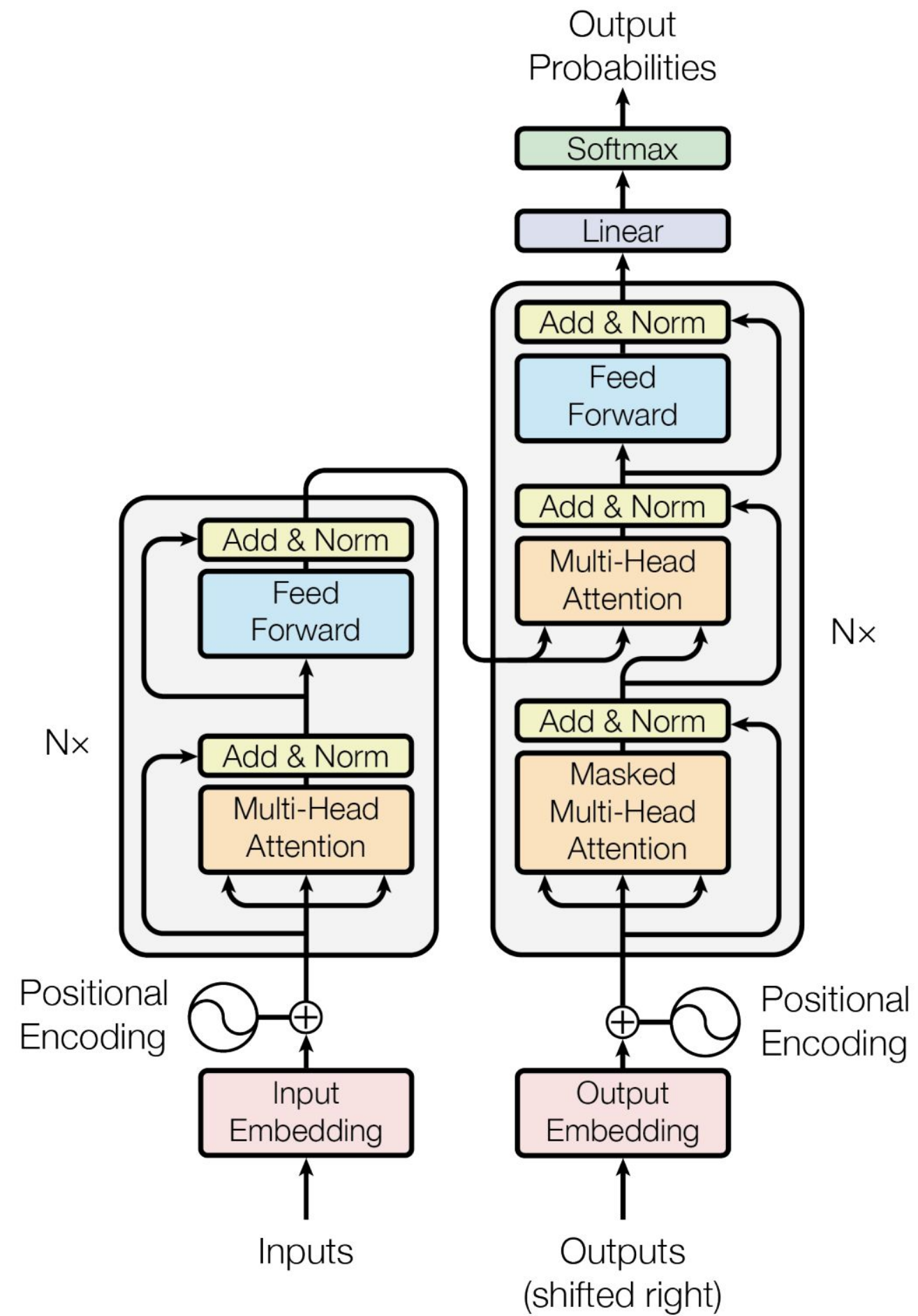


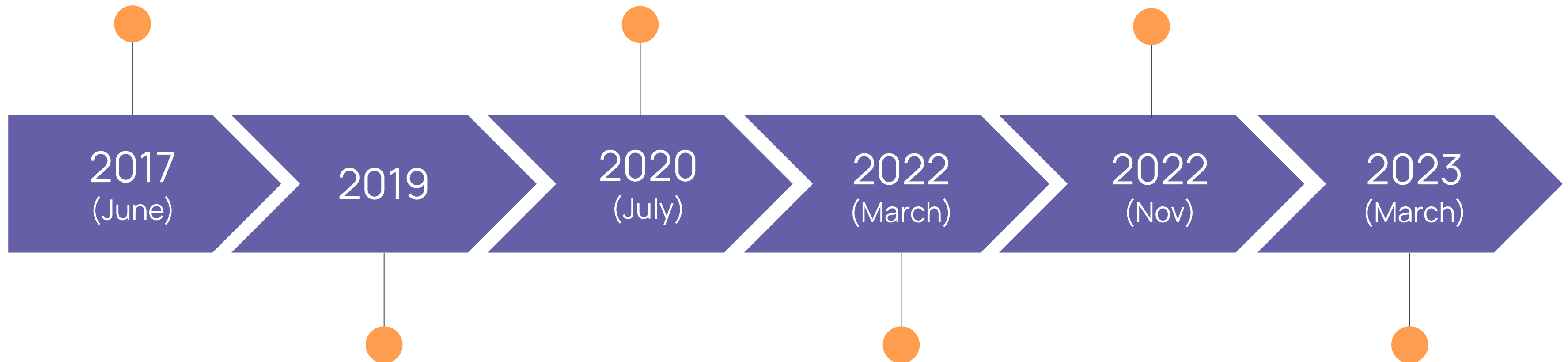
Figure 1: The Transformer - model architecture.



Attention is All You Need
(Transformers)

Language Models are
Few-Shot Learners
(GPT-3)

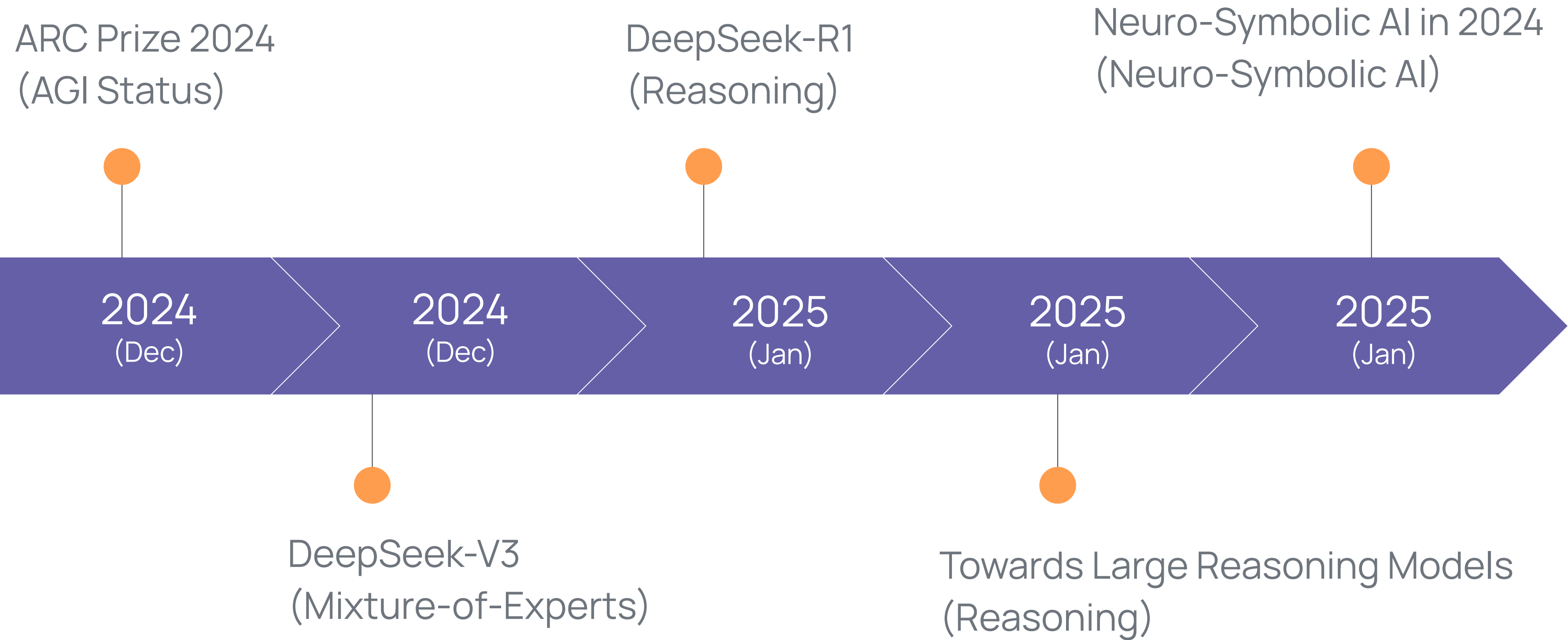
ChatGPT Launched!
(1st Mainstream RLHF-aligned Chatbot)



Language Models are
Unsupervised
Multitask Learners
(GPT-2)

Training language models to
follow instructions with
human feedback
(RLHF for human-alignment)

GPT-4 Technical Report
(Multimodal inputs)





Gemini



Prompt Engineering Elements

Instruction - a specific task or instruction you want the model to perform

Context - external information or additional context that can steer the model to better responses

Input Data - the input or question that we are interested to find a response for

Output Indicator - the type or format of the output.



Text-to-Text



How can I help you today?

Come up with concepts
for a retro-style arcade game

Suggest some codenames
for a project introducing flexible work arrangements

Make up a story
about Sharky, a tooth-brushing shark superhero

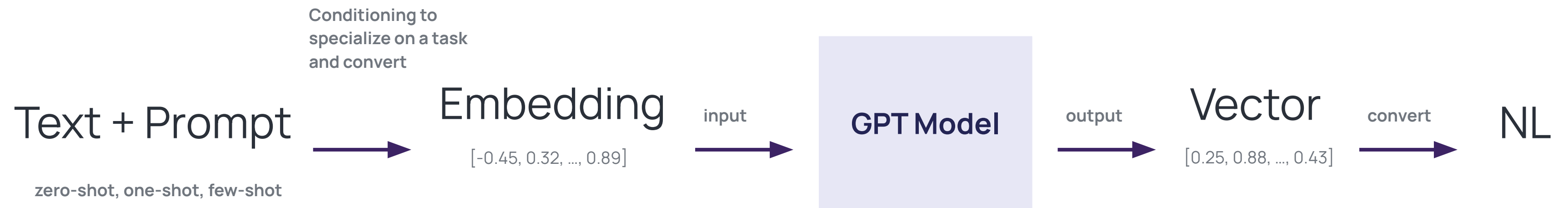
Plan an itinerary
for a fashion-focused exploration of Paris



Message ChatGPT...



Text-to-Text Decoder Models in Action



Learn & Understand

Complex concepts, fields of study, professional development, ...



Research

Market, competition, trends, ...



Write

Emails, reports, thought leadership, ...



Generate

Lists, outlines, surveys, ...



Compare Options

X and Y, X vs Y, X-Y analysis, ...



Create a Plan

Strategic, project, event ...



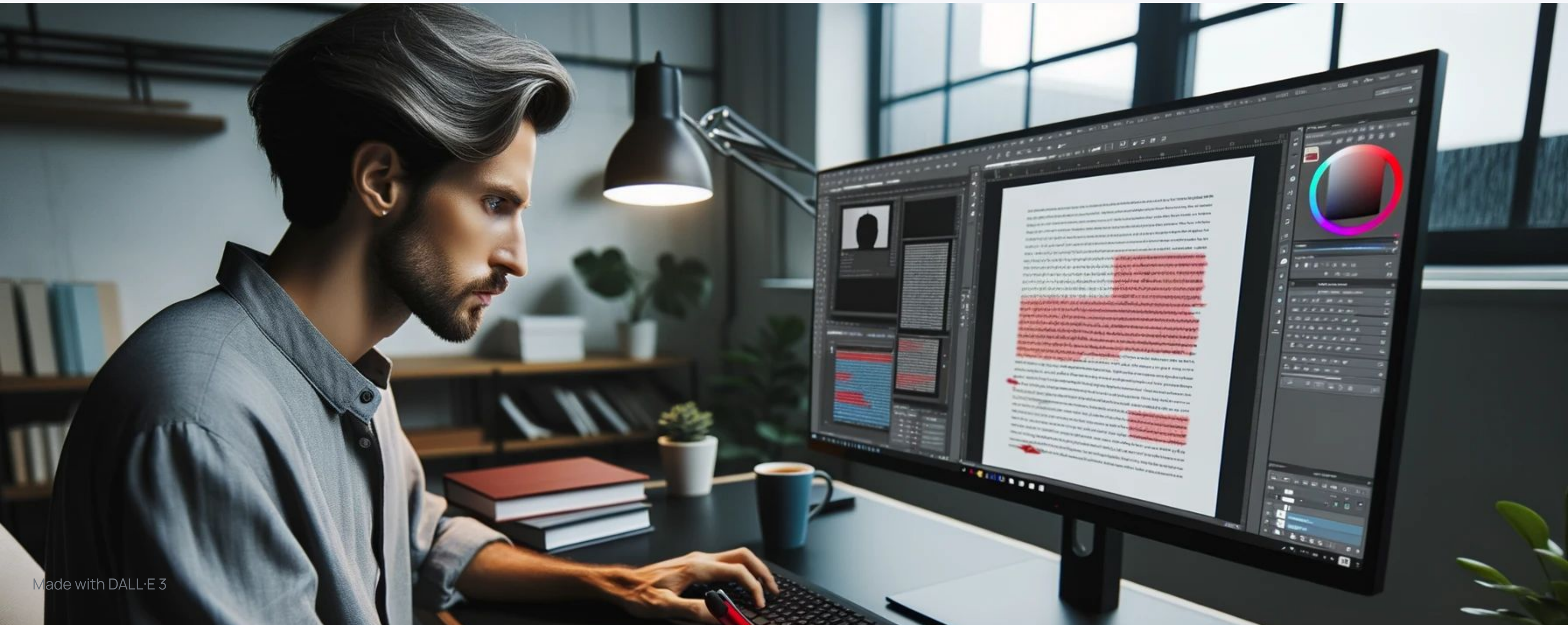
Brainstorm & Ideate

Topics, titles, methods, ...



Edit & Improve

Grammar, wording, effectiveness, ...



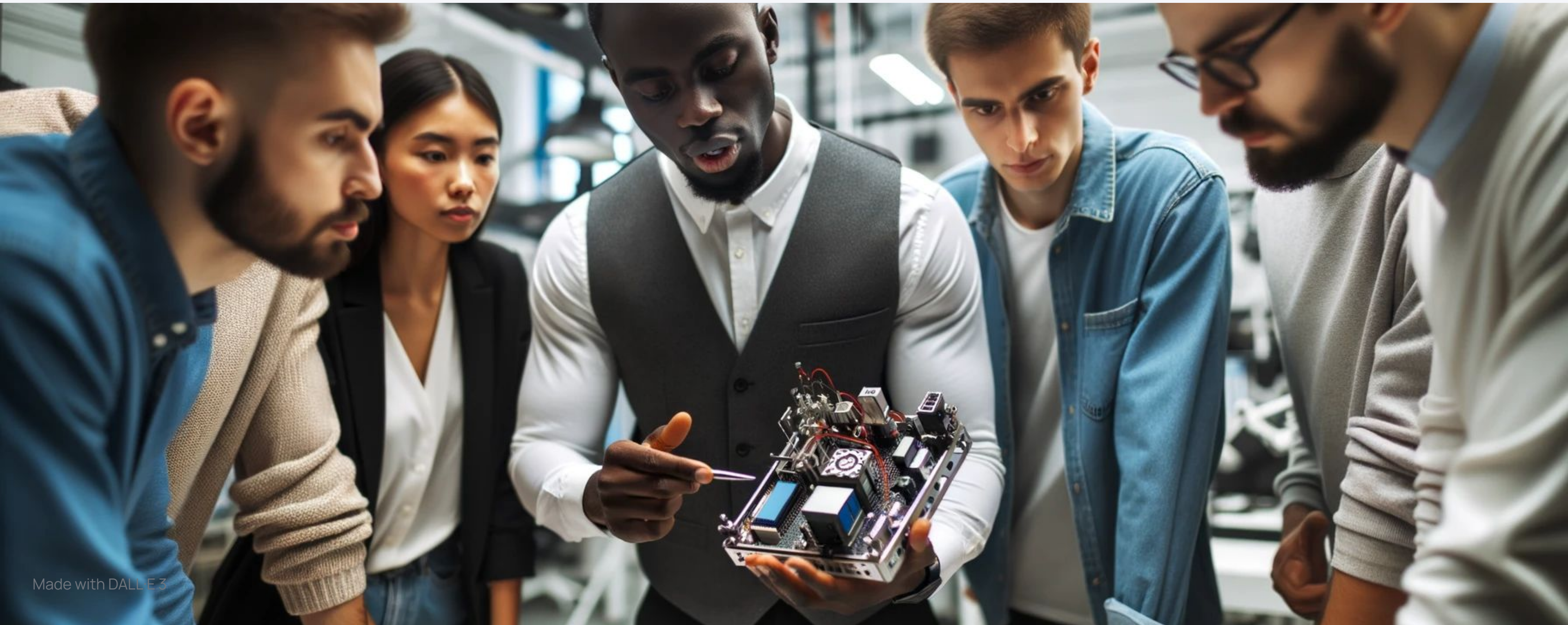
Summarize

Research, content, documents, ...



Explain & Simplify

Complex concepts, topics, how things work, ...



Analyze

Key takeaways, insights, sentiment, ...



Text-to-Code

* Welcome to Claude Code

> Think harder...

```
while(curious) {  
    question everything();  
    dig_deeper();  
    connect_dots(unexpected);  
  
    if (stuck) {  
        keep_thinking();  
    }  
}
```

PyTorch MNIST Experiments

Add mixed precision training, learning rate scheduling, proper validation. Also create an experiment configuration so I can easily run different hyperparameter settings.

Thought 10s

Searched PyTorch mixed precision training best practices
Read notebooks/train_model.py (current implementation)

I'll enhance your MNIST trainer with a complete experiment framework including mixed precision, validation splitting, proper configuration management. Let me start by reviewing the training module:

📄 train_model.py +156 -34

Now let me update the evaluation module to save results and generate a detailed report:

📄 evaluation.py +24 -6

Let me create a command-line runner script to make experiments easy to execute:

Plan, search, build anything...

∞ Agent GPT-5.2 ▾

A woman with long dark hair, wearing a red shirt, is seen from the side, focused on her work. She is sitting at a desk and typing on a silver laptop. The laptop screen displays a code editor with a dark theme and syntax-highlighted text. A large, bold, black '10x' is superimposed over the center of the image, partially covering the woman's face and the laptop screen. The background is a bright, out-of-focus office space with large windows and modern furniture.

10x



Multimodal Open Language Model built by Ai2

Learn about Molmo



What kind of music is this?



Write a story about what is happening here. End t...



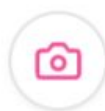
Point to all the faders.



Describe this image.



Count the boats



Add an image and ask Molmo...



○ Upload an image ○ Type a question

Demo

Intelligence Exhibited by Machines

Goals:

- Demos of creative generation and multi-step task execution
- See how models generate multi-modal content from same text

Links:

- [Gamma](#)

Reflect + Discuss:

- Share your thoughts

What is multimodal AI?



Multimodal machine learning aims to design computer agents with intelligent capabilities such as **understanding, reasoning, and learning** through integrating **multiple communicative modalities**



A **modality** refers to a way in which a **natural phenomenon** is **perceived or expressed**.



[Modality]-to-[Modality]



Gemini

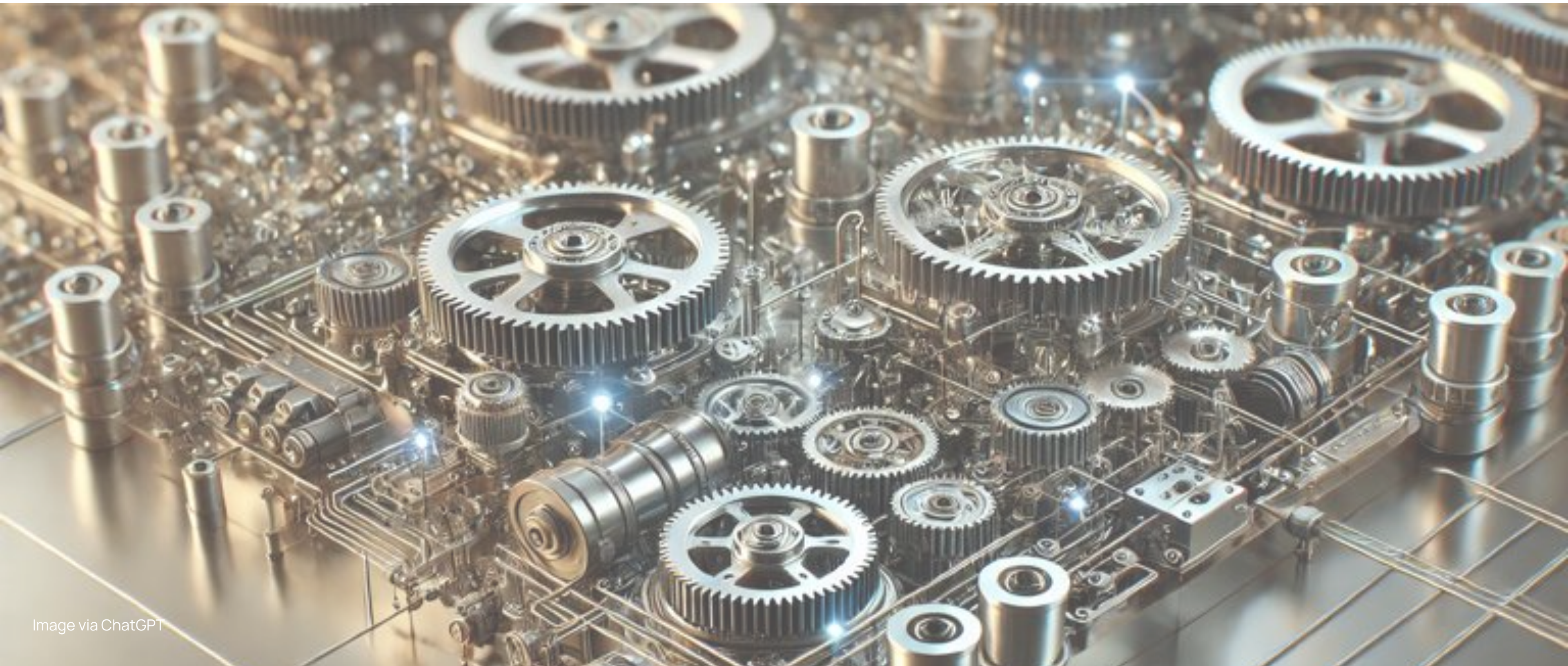




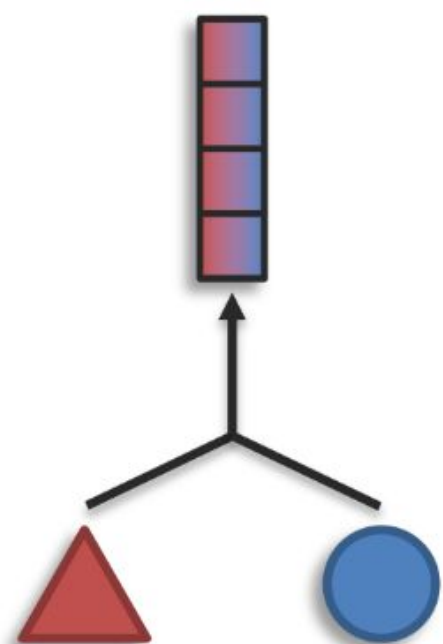
Data $\approx$ Representation



The Inner Workings

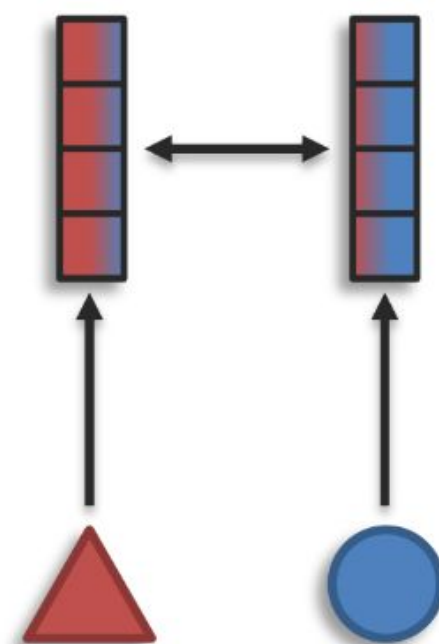


Fusion



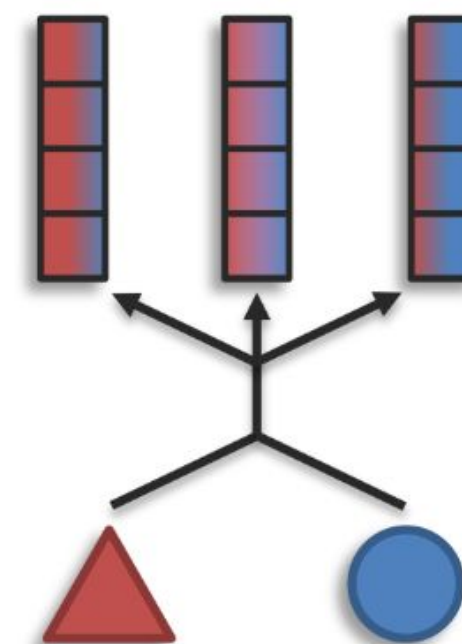
modalities $>$ # representations

Coordination

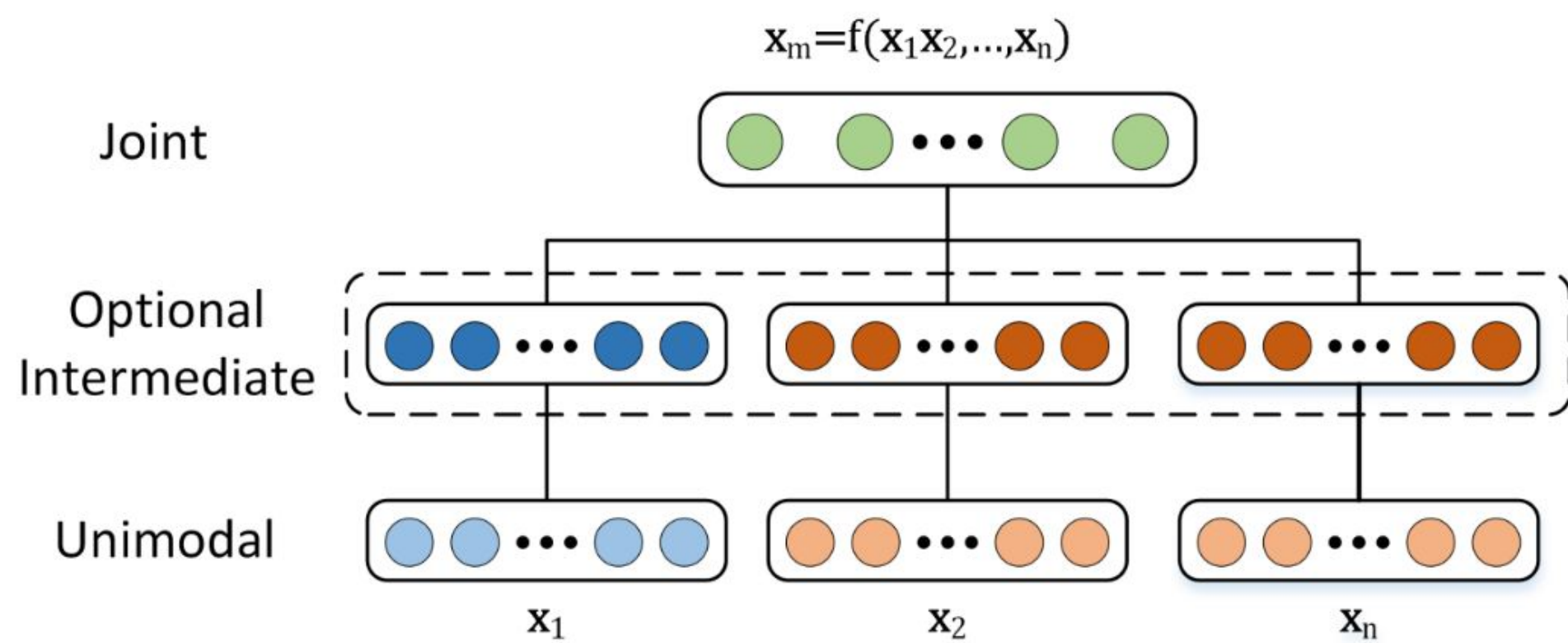


modalities $=$ # representations

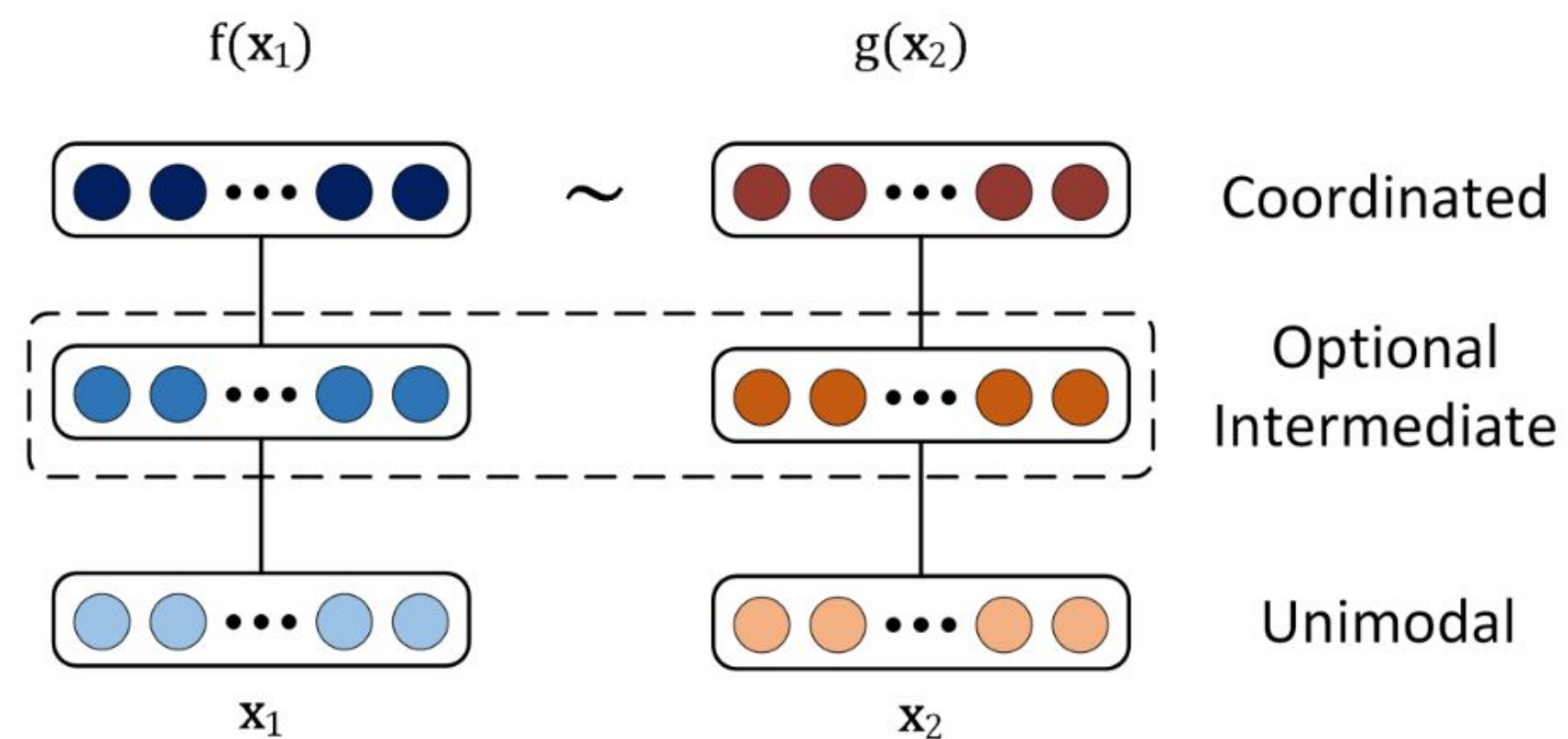
Fission



modalities $<$ # representations

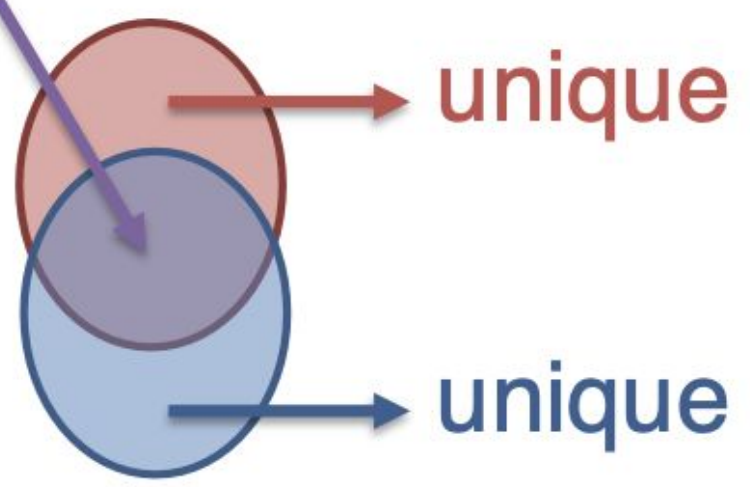


(a) Joint representation

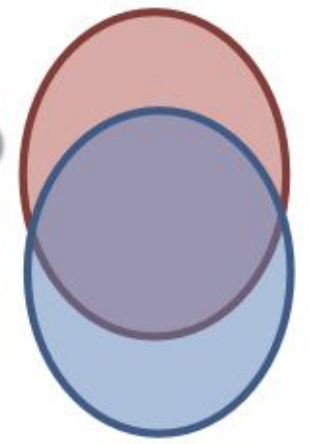


(b) Coordinated representations

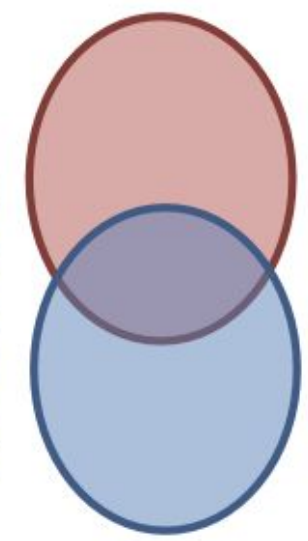
Connections: Shared information that relates modalities



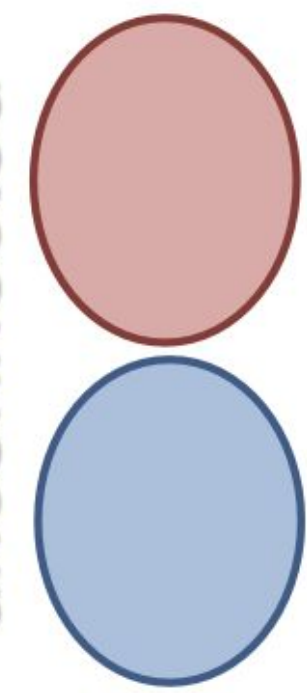
stronger



weaker



unconnected



Statistical

Semantic

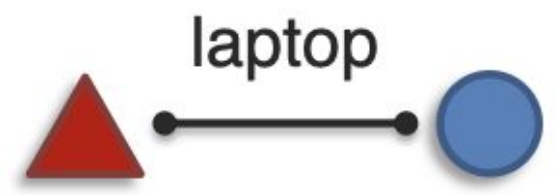


Association

Dependency

Correspondence

Relationship



e.g., correlation,
co-occurrence

e.g., causal,
temporal

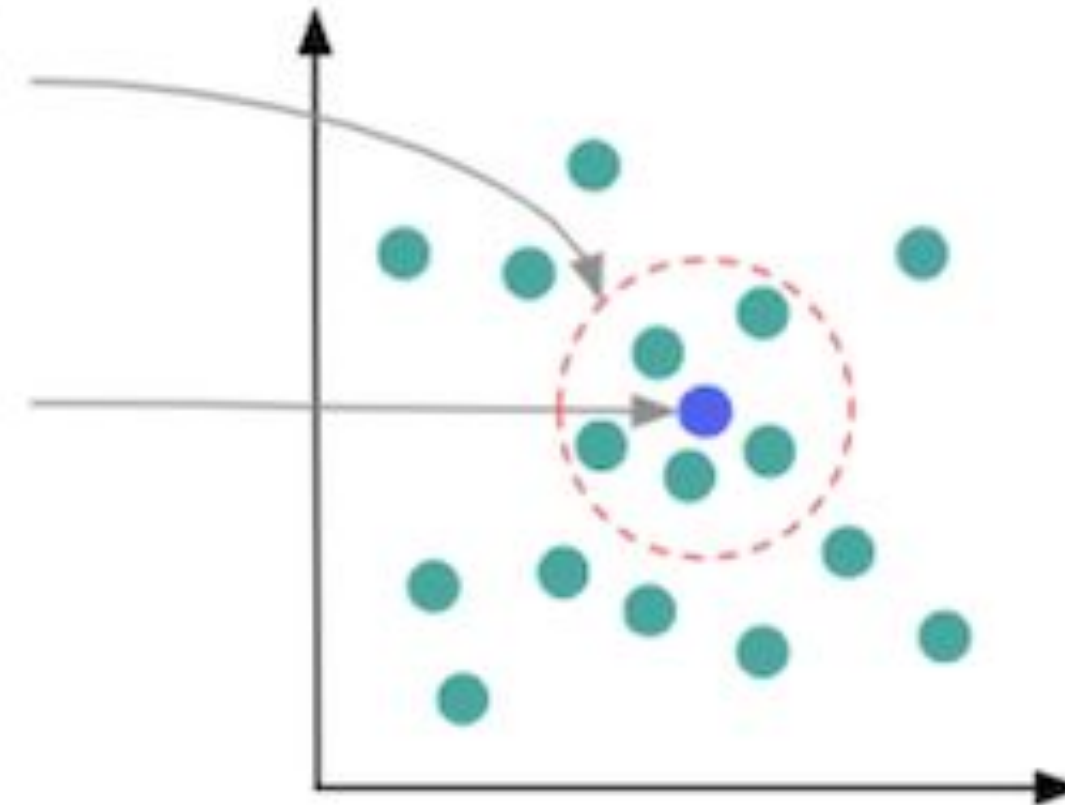
e.g., grounding

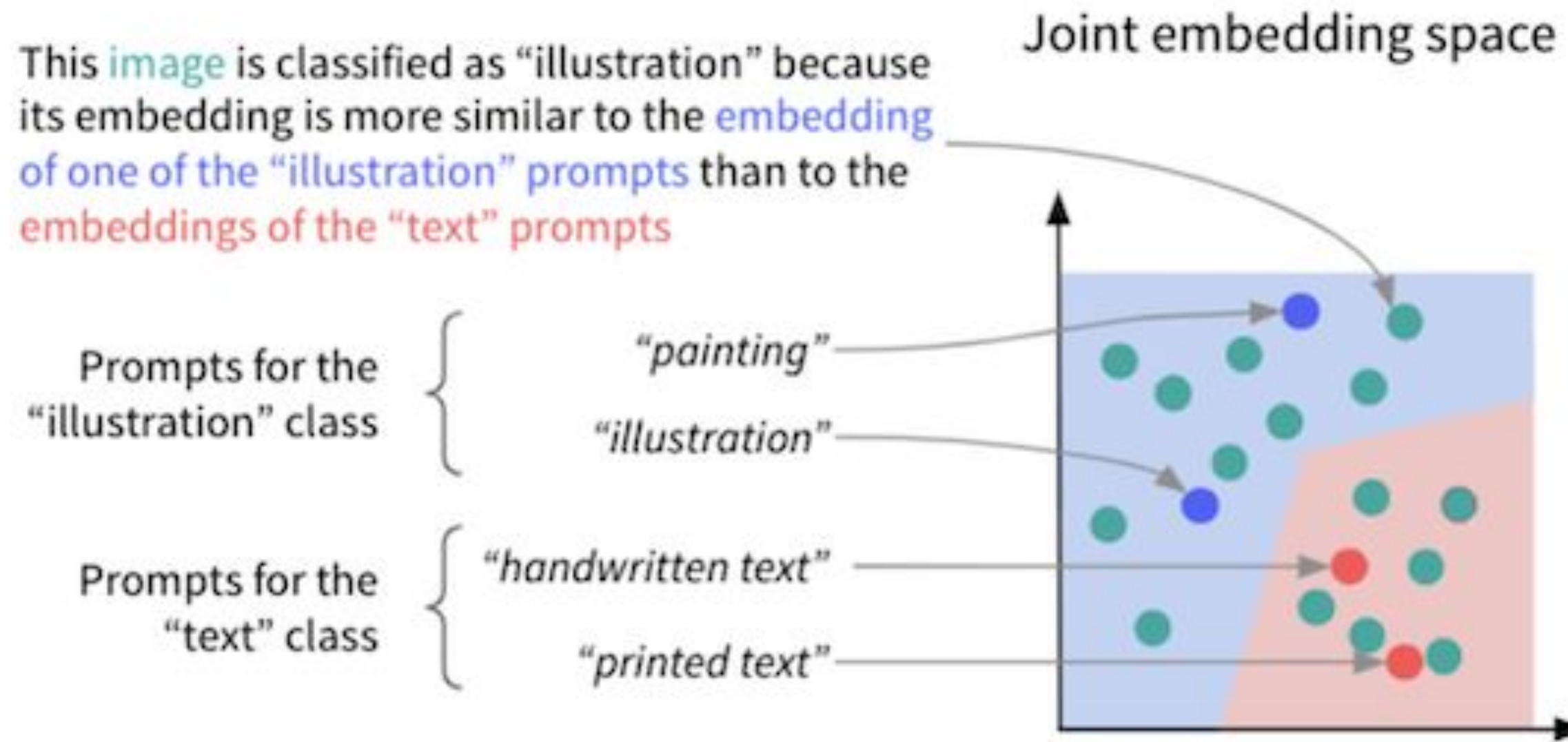
e.g., function

Joint embedding space

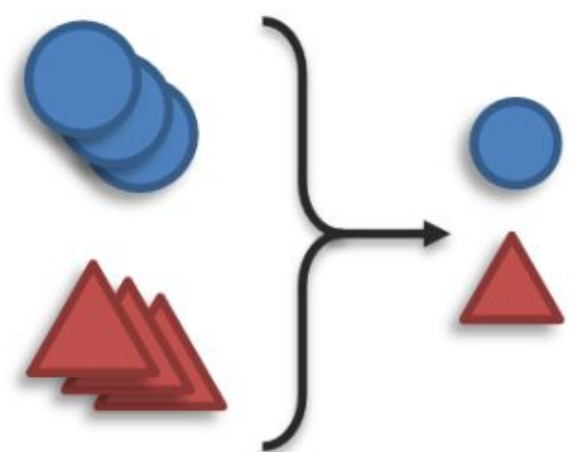
The results of the image search are the **images** whose embeddings are **most similar** to the **query's** embedding

Query: *"painting of a mountain landscape at sunset"*





Summarization



Reduction



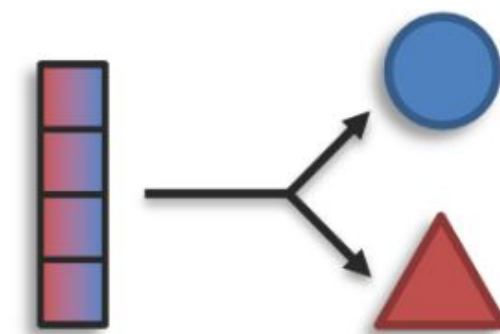
Translation



Maintenance



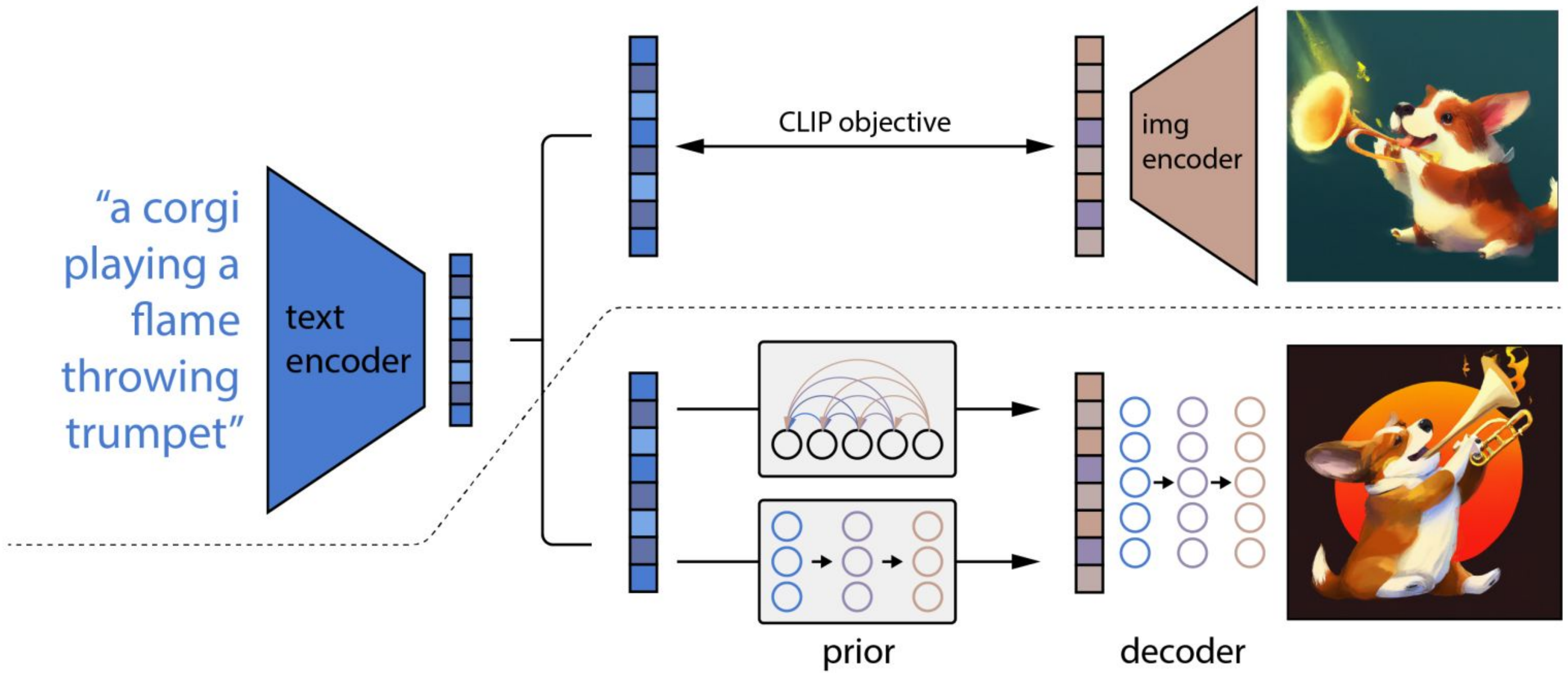
Creation



Expansion



Information:
(content)



Activity

Prompt Clinic (Trustworthy Outputs)

Use Case: Customer support email for a late delivery

Goal: Engineer a prompt that is compliant and drives accuracy, trust, and repeatability

*** Don't use any devices for assistance

*** Follow guidance on activity steps

Activity

Customer message

“Order #48392 due Friday. Now Tuesday. Tracking hasn’t updated. Needed as a gift. Refund shipping or cancel? —Jamie”

Rules (must follow)

- Empathetic. No blame.
- Verifiable only: no ETA/date, no “I checked tracking,” no invented details.
- Offer only: A) expedite replacement, B) cancel + full refund, or C) refund shipping if delivered in 48 hours.
- Format: Subject + 3 short paragraphs + bullets (120–160 words).

Steps (run at least three times each)

1. Baseline: “Write a friendly reply to Jamie.”
2. Your prompt: add Role | Rules | Format | Verifiability guardrail..

*** Don’t use any devices for assistance
*** Follow guidance on activity steps

Activity

Discussion:

- What did the baseline output hallucinate, get wrong, or claim that wasn't supported by the input or facts? What about your prompt?
- Which element mattered most: Role, Rules, Format, or Guardrail, and why?
- What's the highest-risk failure mode here, and what exact prompt line prevents it?
- How would you make this reusable: what are the variables vs the fixed rules?
- What "QA check" would you add before sending an LLM output to a customer?
- How ensure governance and consistency across the org?

*** Don't use any devices for assistance

*** Follow guidance on activity steps